Design and Fabrication of Low-Temperature Continuous-Flow Cryostat System

Nguyen Xuan Tai, Nguyen Tien Dat, Tran Phan Sang, Nguyen Minh Tam

Department of Advanced Materials Science and Nanotechnology, University of Science and Technology of Hanoi (USTH)
Instructors: Dr. Nguyen Van Quyen (AMSN, USTH) • Nguyen Van Huan (An Tai Loc Co., Ltd.)

USTH AMSN
Cryogenic Instrumentation Lab
Group Project 2026

1. MOTIVATION & CRYOGENIC PHYSICS

Cryogenic environments are critical for characterizing low-dimensional semiconductor devices, phase transitions, and quantum transport phenomena. Commercial turn-key cryostats are cost-prohibitive for flexible prototyping and laboratory education.

Core Objective: Develop an accessible, modular continuous-flow cryostat operating between **77 K and 300 K** with high-vacuum thermal decoupling ($P < 10^{(-4)}$ mbar) and in-situ 4-probe electrical transport tracking.

Thermodynamic Heat Load Balance:

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{cond}} + \dot{Q}_{\text{rad}} + \dot{Q}_{\text{conv}}$$

- **Conduction Leakage:** Fourier's law with temperature-dependent conductivity $k(T)$:

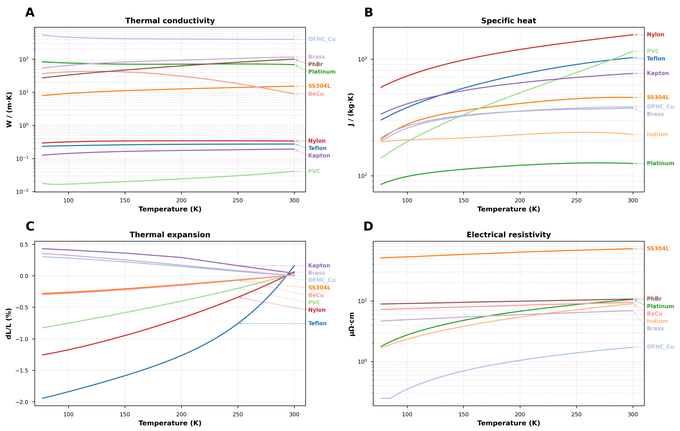
$$\dot{Q}_{\text{cond}} = \frac{A}{L}\int_{T_C}^{T_H} k(T)dT$$

- **Radiation Transfer:** Stefan-Boltzmann relation with low-emissivity radiation shielding:

$$\dot{Q}_{\text{rad}} = \varepsilon\sigma A(T_H^4 - T_C^4)$$

- **Molecular Flow Vacuum Regime:** At $P < 10^{(-4)}$ mbar, the gas molecular mean free path exceeds chamber dimensions: $\dot{Q}_{\text{conv}} \approx 0$.

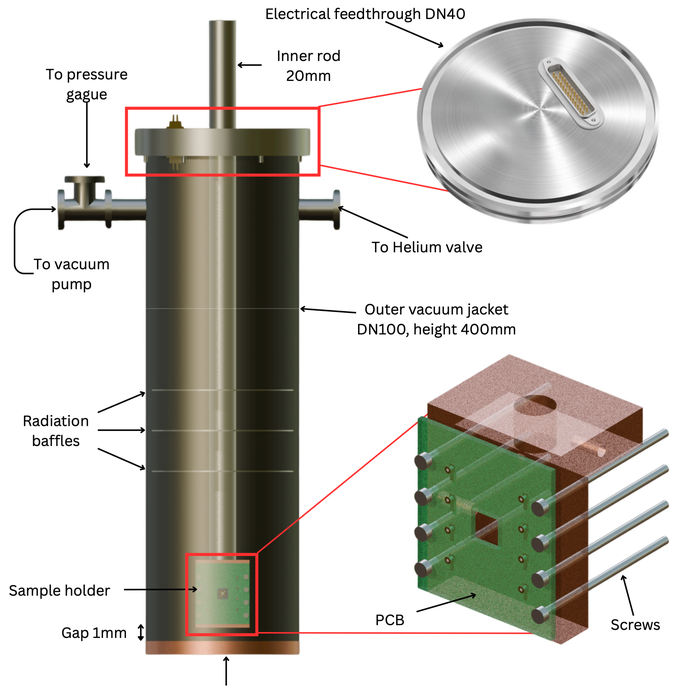
2. MATERIALS & THERMAL COUPLING



Thermal conductivity $k(T)$ and electrical resistance across materials.

- **OFHC Copper (C10100):** Cold finger & sample stage ($k \approx 390\ \text{W/(m·K)}$ at 300 K; $k > 1000\ \text{W/(m·K)}$ around 50 K) ensures rapid thermal equilibrium.
- **SS304L Stainless Steel:** Outer vacuum shell minimizing conductive loss ($k \approx 15\ \text{W/(m·K)}$).
- **Phosphor Bronze / Manganin:** Signal wiring reducing parasitic electrical lead heat load.

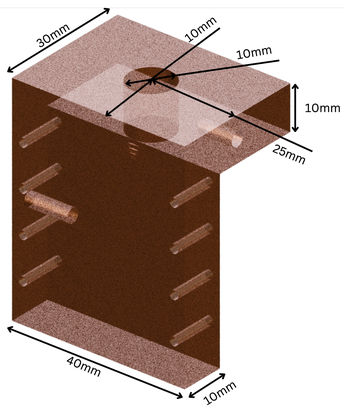
3. MECHANICAL ARCHITECTURE & CAD



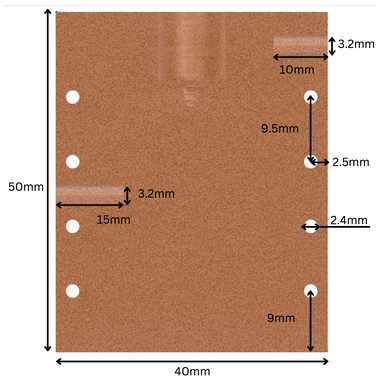
Full 3D cross-sectional CAD assembly of the cryostat.

- **Modular Vacuum Enclosure:** SS304L body with standard KF-40 flanges, viton O-rings, and quick-release centering clamps.
- **Cryogenic Delivery Tube:** Continuous liquid nitrogen delivery tube entering the cold head with minimal parasitic conductive bridges.

4. SAMPLE STAGE & 4-PROBE INTERFACE



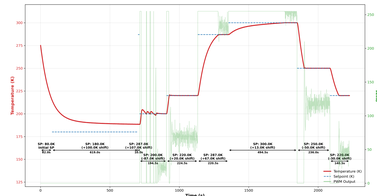
3D OFHC sample stage.



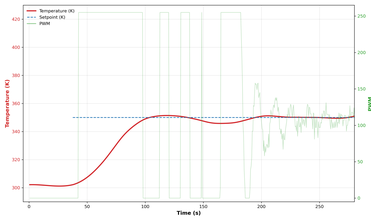
FreeCAD drawing.

- **L-Shaped Geometric Profile:** Optimizes conductive contact area with the cold finger while preserving optical and electrical top access.
- **Electrical Contact Interface:** Gold-plated phosphor-bronze spring probes mounted on a low-outgassing PCB base for 4-probe van der Pauw measurements.

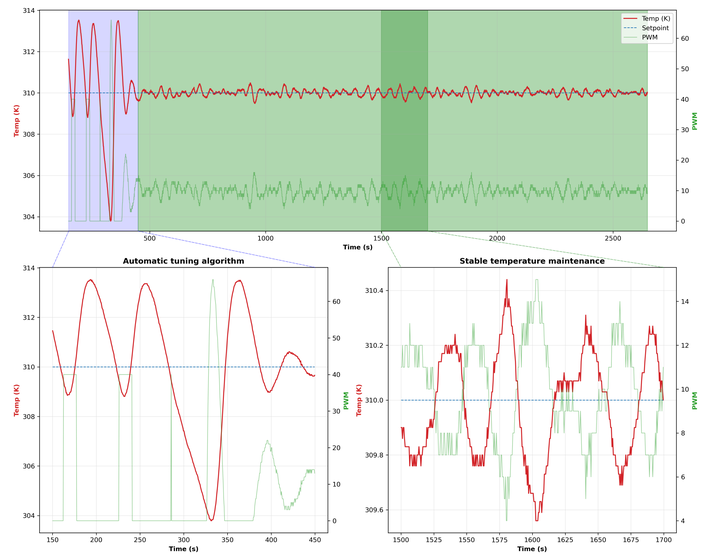
5. SIMULATION & EXPERIMENTAL VALIDATION



Continuous-flow cooldown curve.



Vacuum pump-down curve.



PID thermal regulation (± 0.4 K) across multi-step setpoints.

- **Dynamic Thermal Tracking:** Åström-Hägglund relay auto-tuning with Tyreus-Luyben rules smoothly managed thermal inertia.
- **Thermal Stability:** Maintained setpoints with ± 0.4 K fluctuation from 160 K to 300 K without sustained oscillation.

6. SPECIFICATIONS & CONCLUSION

Parameter	Demonstrated Specification
Temperature Range	160 K – 300 K (Tested) / 77 K (Baseline LN2)
Temperature Stability	± 0.4 K steady-state fluctuation
Base Vacuum	$8.5 \times 10^{(-5)}$ mbar (molecular flow)
Sample Mount	OFHC copper stage with 4-probe spring pins
Control Loop	Arduino PID + Python GUI with live telemetry

Conclusion: Validated a robust, modular, accessible continuous-flow cryostat platform integrating high vacuum isolation, precision PID temperature control, and in-situ electrical resistivity characterization.